M08 Demo Day — Team Evidence & Presentation Outline

AI4I 2020 Predictive Maintenance | Team: Keeshaun Cobb, Brett Harrison, Wynn Nguyen, Zach Lee | October 18, 2026

Repository: https://github.com/37216110/cis480_capstone

Evidence status: Model statistics below are transcribed from Brett's notebook PDF outputs. The Tableau workbook is an exploratory draft; its live connection and final visuals have not been independently verified.

1. Stakeholder need, project scope and report narrative

A hypothetical manufacturing maintenance manager needs evidence about whether operating measurements can help flag machine-failure records for human review. Our MVP is an analytical classification pipeline, not a deployed monitoring system or an advance-time failure forecast. It uses the public, synthetic AI4I 2020 dataset.

The six predictors are machine Type, air temperature, process temperature, rotational speed, torque and tool wear. Product IDs, row IDs and five failure-mode flags are excluded from model inputs to avoid leakage. Brett's cleaning step removed 27 label-inconsistent rows, leaving 9,973 observations (330 failures, 3.31%).

2. Method, working MVP and model comparison

Both models use the same stratified 80/20 train/test split (7,978 training, 1,995 test; random seed 42), five-fold cross-validation on the training partition, and a 0.50 decision threshold for the reported held-out evaluation.

Held-out test measure	Logistic Regression	Random Forest
Accuracy	97.04%	97.89%
Precision	73.33%	65.00%
Recall	16.67%	78.79%
F1	27.16%	71.23%
ROC-AUC	0.9194	0.9725
Average precision	0.4918	0.8030
True negatives	1,925	1,901
False positives	4	28
False negatives	55	14
True positives	11	52

Decision: Random Forest is the stronger research prototype for detecting failures because it detects 41 more actual failures on this test split, although it triggers 24 more false alarms. A model that predicts no failures would achieve approximately 96.69% accuracy yet miss all 66 failures, illustrating why accuracy alone is misleading.

3. Tableau evidence

The provided CIS_480_Rough_Draft.twb contains four worksheets: “Air Temp Vs Process Temp,” “Torque Vs Rotate Speed(Type H),” “Torque Vs Rotate speed(Type L),” and “Torque Vs Rotate speed(Type M).” These are exploratory views of operating measurements; they are not verified failure-rate charts, final dashboards, or model validation evidence.

The workbook references a temporary local Tableau .hyper connection. Before a live demo, Zach or another team member should reconnect it to the cleaned CSV, verify each worksheet’s fields and calculations, save a portable workbook or publish an accessible view, and export screenshots with descriptive captions. If this cannot be completed, present the four worksheet titles as preliminary work only and use Brett’s notebook figures for verified model evidence.

4. Interpretation and stakeholder impact

Wynn’s interpretation identifies rotational speed and torque as the strongest Random Forest predictors, followed by air temperature and tool wear. Brett’s average-precision permutation-importance scores are 0.4496 (rotational speed), 0.4386 (torque), 0.3175 (air temperature), 0.2597 (tool wear), 0.1820 (process temperature) and 0.0448 (machine type). These rankings are model-specific predictive associations, not evidence of causal machine-failure mechanisms.

Stakeholder implication: prioritize review of rotational speed, torque, temperature and tool wear measurements, but require human oversight and independent validation before using model predictions for maintenance decisions.

5. Acceptance-test record

ID / criterion	Expected	Observed / status	Evidence / owner
A1 / cleaning	Reproducible clean dataset with documented exclusions	9,973 rows after 27 exclusions — reported PASS; rerun pending	Brett cleaning utilities and cleaned CSVs / Brett
A2 / leakage control	No ID or failure-mode predictors	Six permitted predictors — reported PASS; code review pending	Model notebooks / Brett
A3 / fair comparison	Same held-out records and threshold	1,995 test cases, 66 failures, threshold 0.50 — reported PASS	Model notebook exports / Brett
A4 / useful failure detection	Compare recall and false alarms to baseline	RF 52/66 detected vs. LR 11/66 — PASS for comparison; no approved recall target	Confusion matrices / Brett
A5 / accessible Tableau	Workbook opens against shareable clean data	NOT VERIFIED: temp .hyper path; draft only	CIS_480_Rough_Draft.twb / Zach
A6 / reproducible handoff	Fresh run from documented environment	NOT VERIFIED in this package	GitHub README and notebooks / team

6. Validation record

Input / question	Expected result	Actual result	Status and evidence
Do 1,995 test records include 66 failures?	1,929 nonfailures and 66 failures	Confusion matrices sum to 1,995 for each model	PASS by arithmetic; notebook PDF tables
Does RF recall equal 52/66?	78.79%	78.79%	PASS by arithmetic; RF notebook PDF
Does LR recall equal 11/66?	16.67%	16.67%	PASS by arithmetic; LR notebook PDF
Is RF more sensitive but less precise?	Higher recall; more false positives	RF: 78.79%, 28 FP; LR: 16.67%, 4 FP	PASS by reported results; notebook PDFs
Can Tableau draft load clean data elsewhere?	All worksheets refresh from cleaned CSV	Not tested; temporary source path identified	PENDING; Tableau .twb

7. Impact-risk register

Risk	Impact / likelihood	Mitigation and owner
Synthetic-to-real transfer	High / unknown	External validation on authorized real-machine data; team
Missed machine failures	High / demonstrated (14 RF false negatives)	Tune threshold using validation data; Brett
False maintenance alarms	Medium / demonstrated (28 RF false positives)	Compare alert burden and maintenance costs; Wynn / Brett
Data imbalance and label inconsistencies	High / demonstrated	Document prevalence and assess sensitivity to 27 exclusions; Brett
Interpretation mistaken for causality	Medium / possible	State importance is associative; Wynn
Tableau broken connection	Medium / identified	Reconnect and package data; Zach
Incomplete team sign-off	Medium / unconfirmed	Circulate final package, document responses, contact instructor if needed; Keeshaun

8. Limitations and next decision

The evidence supports only classification of synthetic failure records on one held-out split. It does not establish real-world generalization, prediction lead time, production safety, causal mechanisms or optimal maintenance cost. The 27 excluded inconsistent records may influence results; a sensitivity analysis is warranted. Permutation importance was calculated on the test set and should be interpreted as descriptive model analysis rather than an independently confirmed feature-ranking validation.

Recommended next decision: keep Random Forest as the candidate prototype, then test threshold selection on validation data, evaluate false-alarm costs, rerun the complete workflow independently, and reconnect Tableau to a portable data source before any operational use.

9. Current work plan and contribution evidence

Owner / task	Status	Dependency	Acceptance criterion / evidence
Brett / cleaning and models	Reported complete	Public AI4I CSV	Scripts, cleaned data, notebooks, test charts in repo
Wynn / interpretation and limitations	Written interpretation supplied	Brett model outputs	Message identifying predictors, recall tradeoff and caveats
Zach / Tableau exploration	Draft supplied	Data connection and cleaned CSV	Four .twb worksheets; final refresh pending
Keeshaun / integrated presentation and documentation	Draft package prepared	Team evidence and review	PPTX, outline, test record, risk register and work plan
All members / review and demo readiness	Not yet confirmed	Final package shared	Each member acknowledges review and can explain method, limitation and own artifact

10. Slide-by-slide accessible text outline

1 — Project introduction: The project tests whether six operating measurements can help identify synthetic machine-failure records. Team roles and prototype scope are shown.

2 — Stakeholder need: A hypothetical maintenance manager needs actionable, inspectable evidence. The tool supports human review; it does not automate maintenance.

3 — Research questions: Compare failure patterns, identify predictive measurements, and compare classification models with a no-failure baseline.

4 — MVP workflow: Public AI4I dataset → clean 27 inconsistent labels → six predictors → stratified 80/20 split → Logistic Regression and Random Forest → metrics and interpretation.

5 — Preliminary Tableau: Four draft worksheets: air versus process temperature, and torque versus rotational speed for H, L and M machine types. Draft uses a temporary data connection; no validated failure-rate claim.

6 — Logistic Regression: Held-out recall 16.67%; TP 11, FN 55, FP 4, TN 1,925. Accuracy 97.04% can obscure missed failures.

7 — Random Forest: Held-out recall 78.79%; TP 52, FN 14, FP 28, TN 1,901. Stronger failure detection with additional false alarms.

8 — Feature importance: Permutation importance ranks rotational speed and torque first, followed by air temperature and tool wear. Associations are not causal.

9 — Supported claims and limits: Evidence supports a test-set comparison on synthetic data. It does not establish real-world deployment, advance warning, or live Tableau reproducibility.

10 — Handoff and next steps: Credit each member's evidence; reconnect Tableau, rerun notebooks, test thresholds, and document team review.

11. Sources and artifact links

Team repository: https://github.com/37216110/cis480_capstone

AI4I 2020 Predictive Maintenance Dataset, UCI Machine Learning Repository: https://archive.ics.uci.edu/dataset/601/ai4i+2020+predictive+maintenance+dataset

Local evidence used to assemble package: Logistic_regression.pdf; Random_forest.pdf; CIS_480_Rough_Draft.twb

12. AI-use disclosure and verification

OpenAI ChatGPT assisted with organizing the presentation, rewriting explanatory text, preparing the acceptance/validation tables and risk register, and checking arithmetic against the model outputs. Numerical claims were checked against the team's provided notebook PDF outputs and confusion-matrix arithmetic; a fresh end-to-end notebook execution and live Tableau refresh have not been performed in this package.